Detailed Step-by-Step Proofs: Rubbo (2024)

Networks, Phillips Curves, and Monetary Policy

+ Code Walkthrough + Small Open Economy Extension

Working Notes

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1 Model Setup and Notation

1.1 Household

The representative household maximizes:

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \rho^t \left[\frac{C_t^{1-\gamma}}{1-\gamma} - \frac{L_t^{1+\varphi}}{1+\varphi} \right] \tag{1}$$

subject to the budget constraint:

$$P_t^C C_t + B_{t+1} \leq W_t L_t + \Pi_t - T_t + (1+i_t) B_t \tag{2}$$

Optimality conditions:

- **Labor supply (intratemporal):** $W_t / P_t^C = C_t^\gamma L_t^\varphi$ — the wage equals the marginal rate of substitution between consumption and leisure.
- **Euler equation (intertemporal):** $(1+i_{t+1}) = \rho^{-1} \mathbb{E}_t \left[(C_{t+1}/C_t)^\gamma (P_{t+1}^C / P_t^C) \right]^{-1}$.

1.2 Producers

There are N industries. Industry i has a unit mass of monopolistically competitive firms. Firm f in sector i produces:

$$Y_{if} = A_i F_i\big(L_{if}, \{X_{ijf}\}_{j=1}^N\big) \tag{3}$$

where A_i is TFP and F_i is **constant returns to scale** (CRS). All firms in sector i share the same production technology.

Demand for firm f's output comes from households and other producers:

$$Y_{if} = \left(\frac{P_{if}}{P_i} \right)^{-\varepsilon_i} Y_i \tag{4}$$

where $\varepsilon_i > 1$ is the within-sector elasticity of substitution and $P_i = \left(\int_0^1 P_{if}^{1-\varepsilon_i} df \right)^{1/(1-\varepsilon_i)}$ is the sector i price index.

Cost minimization by firm f gives industry-level marginal cost (independent of output by CRS):

$$MC_{it} = \min_{\{L_{if}\},\{X_{ijf}\}} \frac{W_t L_{if} + \sum_j P_{jt} X_{ijf}}{Y_{if}} = c_i(W_t, \{P_{jt}\}, A_{it}) \tag{5}$$

Calvo price setting: Each period, fraction δ_i of firms in sector i get to reset price. The optimal reset price satisfies:

$$P_{it}^* = \frac{\varepsilon_i}{\varepsilon_i - 1} \cdot \frac{\mathbb{E} \sum_{s=0}^{\infty} \rho^s (1-\delta_i)^s Y_{it+s}(P_{it}^*)\, MC_{it+s}}{\mathbb{E} \sum_{s=0}^{\infty} \rho^s (1-\delta_i)^s Y_{it+s}(P_{it}^*)/P_{it+s}} \tag{6}$$

This is a weighted average of current and future marginal costs, with weights declining in $(1-\delta_i)^s$ because with probability $(1-\delta_i)^s$ the price will still be in effect s periods later.

1.3 Key Notation Reference

Symbol	Definition	Matlab variable
$\tilde{y}_t \equiv y_t - y_{t,\text{nat}}$	Aggregate output gap	`IRy`
$\boldsymbol{\pi}_t = (\pi_{1t}, \dots, \pi_{Nt})^T$	Sectoral inflation	`IRpi`
$\boldsymbol{\mu}_t = (\mu_{1t}, \dots, \mu_{Nt})^T$	Log-markups	`mu`
$\boldsymbol{\beta} \in \mathbb{R}^N$	Consumption shares	`beta`
$\boldsymbol{\alpha} \in \mathbb{R}^N$	Labor shares	`alpha`
$\Omega \in \mathbb{R}^{N\times N}$	Input-output matrix	`Omega`
$(I-\Omega)^{-1}$	Leontief inverse	`(eye(n)-Omega)^(-1)`
$\boldsymbol{\lambda}^T = \boldsymbol{\beta}^T (I-\Omega)^{-1}$	Domar weights	`lambda`
$\hat{\delta}_i \equiv \frac{\delta_i(1-\rho(1-\delta_i))}{1-\rho\delta_i(1-\delta_i)}$	Effective Calvo param.	`Delta` (diagonal)
$(I-\Omega\Delta)^{-1}$	Rigidity-adj. Leontief	`(eye(n)-Omega*Delta)^(-1)`
$\mathcal{B}$	Phillips curve slope vector	`b`
$\mathcal{V}$	Cost-push matrix	`v`

Key intuition on the IO matrix Ω: ω_{ij} = fraction of sector i's total costs that come from buying sector j's output. So Ω_{ij} measures how important sector j is as a direct input supplier to sector i.

The **Leontief inverse** $(I-\Omega)^{-1}$ sums up all indirect linkages: $(I-\Omega)^{-1} =$

$I + \Omega + \Omega^2 + \Omega^3 + \cdots$ where Ω^n_{ij} = total flow from j to i through chains of length exactly n. So sector i's cost ultimately depends on sector j's price through *all* indirect paths.

Domar weights $\lambda_i = \beta^T(I - \Omega)^{-1}_i$ = sector i's total sales share in GDP, counting both direct sales to consumers and indirect sales through other producers. Always $\sum_i \lambda_i > 1$.

2 Log-Linearized Building Blocks

2.1 Log-Linearized Marginal Cost

Starting from the cost function $MC_{it} = c_i(W_t, \{P_{jt}\}, A_{it})$, log-linearize around steady state (where all variables are at their ergodic means). Use lowercase for log-deviations: $mc_{it} = \log MC_{it} - \log \overline{MC}_i$, etc.

By Shephard's lemma, the cost-minimizing labor demand is $L_{it} = \partial MC_{it}/\partial W_t$, so its cost share is $\alpha_i = W_t L_{it}/(MC_{it} Y_{it})$. Similarly, $\omega_{ij} = P_{jt} X_{ijt}/(MC_{it} Y_{it})$.

Log-linearizing:

$$\boldsymbol{mc}_t = \boldsymbol{\alpha}\, w_t + \Omega\, \boldsymbol{p}_t - \log \mathbf{A}_t \tag{7}$$

- $\boldsymbol{\alpha} w_t$: higher wages raise costs proportional to labor share
- $\Omega \boldsymbol{p}_t$: higher input prices raise costs proportional to input shares
- $-\log \mathbf{A}_t$: higher productivity lowers costs

2.2 Sector Price Level Dynamics

Because each period a fraction $(1 - \delta_i)$ of firms keep their old price and fraction δ_i reset:

$$P_{it}^{1-\varepsilon_i} = (1 - \delta_i) P_{it-1}^{1-\varepsilon_i} + \delta_i (P^*_{it})^{1-\varepsilon_i} \tag{8}$$

Log-linearizing around zero inflation steady state (where $P^* = P = \bar{P}$):

$$p_{it} = (1 - \delta_i) p_{it-1} + \delta_i p^*_{it} \quad \Rightarrow \quad p^*_{it} - p_{it-1} = \frac{1}{\delta_i}(p_{it} - p_{it-1}) = \frac{\pi_{it}}{\delta_i} \tag{9}$$

2.3 Calvo Optimal Reset Price (Log-Linearized)

Log-linearizing (6) around the zero-inflation, zero-markup steady state:

$$p_{it}^* = (1-\rho(1-\delta_i))\sum_{s=0}^{\infty}[\rho(1-\delta_i)]^s\mathbb{E}_t mc_{it+s} \tag{10}$$

This says: the reset price equals a weighted sum of expected future marginal costs. The weight on period $t+s$ is $[\rho(1-\delta_i)]^s$, declining in the probability $(1-\delta_i)^s$ that the price is still in use.

Subtracting p_{it-1} and using (9):

$$\begin{aligned}\frac{\pi_{it}}{\delta_i} &= (1-\rho(1-\delta_i))\sum_{s=0}^{\infty}[\rho(1-\delta_i)]^s\mathbb{E}_t mc_{it+s} - p_{it-1} \\ &= (1-\rho(1-\delta_i))(mc_{it}-p_{it-1}) + \rho(1-\delta_i)\frac{\pi_{it+1}}{\delta_i}\end{aligned} \tag{11}$$

where the second line uses $p_{it-1} = mc_{it} - (mc_{it} - p_{it-1})$ and iterates forward once.

Rearranging (11) to solve for π_{it}:

$$\pi_{it} = \hat{\delta}_i(mc_{it}-p_{it-1}) + \rho(1-\hat{\delta}_i)\mathbb{E}_t\pi_{it+1} \tag{12}$$

where $\hat{\delta}_i \equiv \frac{\delta_i(1-\rho(1-\delta_i))}{1-\rho\delta_i(1-\delta_i)}$ is the **effective Calvo parameter**. In matrix form with $\Delta = \mathrm{diag}(\hat{\delta}_1,\ldots,\hat{\delta}_N)$:

$$\boldsymbol{\pi}_t = \Delta(\boldsymbol{mc}_t - \boldsymbol{p}_{t-1}) + \rho(I-\Delta)\,\mathbb{E}\boldsymbol{\pi}_{t+1} \tag{13}$$

2.4 Real Wage Equation

Log-linearize the labor supply condition $W_t/P_t^C = C_t^{\gamma}L_t^{\varphi}$ and the CPI $p_t^C = \boldsymbol{\beta}^T\boldsymbol{p}_t$:

$$w_t - \boldsymbol{\beta}^T\boldsymbol{p}_t = \gamma c_t + \varphi l_t \tag{14}$$

Since in the log-linearized model, $c_t \approx y_t$ (goods market clearing) and using Lemmas 1 and 2 below to express things in terms of $\tilde{y}_t$ and $\log\mathbf{A}_t$:

$$w_t - \boldsymbol{\beta}^T\boldsymbol{p}_t = (\gamma+\varphi)\,\tilde{y}_t + \boldsymbol{\lambda}^T\log\mathbf{A}_t \tag{15}$$

3 Proof of Lemma 1: Natural Output = Domar-Weighted TFP

Lemma 1. *In the log-linearized model, natural output (the flex-price equilibrium) satisfies:*

$$y_{t,\text{nat}} = \frac{1+\varphi}{\gamma+\varphi}\,\boldsymbol{\lambda}^T \log \mathbf{A}_t \tag{16}$$

Natural output equals aggregate TFP weighted by Domar sales weights λ_i.

Proof. The flexible-price equilibrium is efficient (a subsidy removes steady-state markup distortions). It solves the social planner's problem:

$$\max_{\{L_i\},\{Y_i\},\{X_{ij}\}} \frac{\mathcal{C}(\{Y_i - \sum_j X_{ji}\})^{1-\gamma}}{1-\gamma} - \frac{(\sum_i L_i)^{1+\varphi}}{1+\varphi} \quad \text{s.t.} \quad Y_i = A_i F_i(\{X_{ij}\}, L_i) \quad \forall i \tag{17}$$

Step 1: Decompose the problem.

Define the *inner problem*: given total labor $L = \sum_i L_i$, allocate it optimally across sectors. This defines:

$$C^*(L;\mathbf{A}) \equiv \max_{\{L_i\},\{Y_i\},\{X_{ij}\}:\sum_i L_i = L} \mathcal{C}(\{Y_i - \textstyle\sum_j X_{ji}\}) \tag{18}$$

By CRS of all F_i and of $\mathcal{C}$, the function $C^*(L;\mathbf{A})$ is **homogeneous of degree 1 in L**: doubling total labor doubles total consumption in the efficient allocation.

The *outer problem*: choose L to maximize $U(C^*(L;\mathbf{A}), L)$. The first-order condition is:

$$C^*(L;\mathbf{A})^{-\gamma} \cdot \frac{\partial C^*}{\partial L} = L^{\varphi} \tag{19}$$

(marginal utility of consumption times marginal product of labor = marginal disutility of labor).

Step 2: Compute $\frac{\partial C^*}{\partial L}$ using the envelope theorem.

The inner problem's Lagrangian is:

$$\mathcal{L} = \mathcal{C}(\{C_i\}) + \sum_i \nu_i [A_i F_i(\{X_{ij}\}, L_i) - Y_i] + \nu_L [L - \sum_i L_i] \tag{20}$$

where $C_i = Y_i - \sum_j X_{ji}$ is final consumption of good i, ν_i is the shadow value of good i, and ν_L is the shadow value of labor.

By the **envelope theorem**: the derivative of the value function with respect to the parameter L equals the partial derivative of the Lagrangian with respect to L (holding all other variables fixed at their optimal values):

$$\frac{\partial C^*}{\partial L} = \nu_L \tag{21}$$

Substituting into (19): $C^{*-\gamma}\nu_L = L^\varphi$, so $\nu_L = C^{*\gamma}L^\varphi$.

Step 3: Compute $\frac{\partial \log C^*}{\partial \log A_i}$ using the envelope theorem again.

Apply the envelope theorem now with respect to A_i:

$$\frac{\partial C^*}{\partial A_i} = \nu_i \frac{\partial [A_i F_i]}{\partial A_i} = \nu_i F_i(\{X_{ij}\}^*, L_i^*) \tag{22}$$

Now convert to logs. Multiply both sides by A_i/C^* and use $A_i F_i = Y_i^*$ (optimal output):

$$\frac{\partial \log C^*}{\partial \log A_i} = \frac{\nu_i Y_i^*}{C^*} \tag{23}$$

Step 4: Identify $\nu_i Y_i^*/C^*$ as the Domar weight λ_i.

In competitive equilibrium, the shadow price ν_i (marginal value of good i) equals its market price scaled by marginal utility: $\nu_i = P_i/(P^C C^{*\gamma})$. Therefore:

$$\frac{\nu_i Y_i^*}{C^*} = \frac{P_i Y_i^*}{P^C C^{*\gamma} C^*} = \frac{P_i Y_i^*}{P^C C^*} = \lambda_i \tag{24}$$

where we used that in the efficient equilibrium $C^{*\gamma} = U_C/\rho$ (normalization) and $P_i Y_i^*/(P^C C^*) = \lambda_i$ is sector i's **Domar weight** (total sales share in nominal GDP).

So from (23): $\frac{\partial \log C^*}{\partial \log A_i} = \lambda_i$.

Step 5: Compute natural employment.

From (19) and (21): $C^{*-\gamma}\nu_L = L^\varphi$, and using $\nu_L = \partial C^*/\partial L$. Since C^* is homogeneous of degree 1 in L, Euler's theorem gives $C^* = L \cdot \partial C^*/\partial L = L\nu_L$. Therefore:

$$C^{*-\gamma}\nu_L = L^\varphi \quad \Rightarrow \quad (L\nu_L)^{-\gamma}\nu_L = L^\varphi \quad \Rightarrow \quad \nu_L^{1-\gamma}L^{-\gamma} = L^\varphi \tag{25}$$

Differentiating the outer FOC with respect to $\log A_i$ (noting $\nu_L = C^{*\gamma}L^\varphi$):

$$\frac{\partial \log \nu_L}{\partial \log A_i} = \gamma\lambda_i + \varphi\frac{\partial \log L}{\partial \log A_i} \tag{26}$$

Also, from the inner problem's labor FOC: $\nu_L = \nu_i \partial(A_i F_i)/\partial L_i$, so $\frac{\partial \log \nu_L}{\partial \log A_i} = \lambda_i + (1-\lambda_i)\frac{\partial \log \nu_L}{\partial \log A_j}\Big|_{j \neq i}$. After iterating and using the budget constraint, one can show $\frac{\partial \log \nu_L}{\partial \log A_i} = (1+\varphi)\lambda_i$. Substituting into (26):

$$(1+\varphi)\lambda_i = \gamma\lambda_i + \varphi\frac{\partial \log L_{\text{nat}}}{\partial \log A_i} \quad \Rightarrow \quad \frac{\partial \log L_{\text{nat}}}{\partial \log A_i} = \frac{1+\varphi-\gamma}{\varphi} \cdot \frac{\lambda_i}{?} \tag{27}$$

Step 6 (direct approach): Use the FOC structure directly.

The key identity is: from Step 3, $\partial \log C^*/\partial \log A_i = \lambda_i$. Since C^* is homogeneous degree 1 in L, we can write $\log C^* = \log L + f(\mathbf{A})$ in the neighborhood of the optimum (this is exact for Cobb-Douglas, and holds to first order generally). Therefore:

$$\frac{\partial \log y_{\text{nat}}}{\partial \log A_i} = \lambda_i + \frac{\partial \log L_{\text{nat}}}{\partial \log A_i} \tag{28}$$

From the outer FOC (19): $\log C^* - \gamma \log C^* = \log \nu_L - \log C^* + \varphi \log L$, or equivalently $(1-\gamma)\log y_{\text{nat}} \approx \text{const} + \varphi \log L_{\text{nat}}$. Differentiating with respect to $\log A_i$:

$$(1-\gamma)\lambda_i + (1-\gamma)\frac{\partial \log L_{\text{nat}}}{\partial \log A_i} = \varphi\frac{\partial \log L_{\text{nat}}}{\partial \log A_i} \tag{29}$$

is not right. Let me use the correct approach via the labor market:

From the labor supply FOC in the natural equilibrium: $\gamma c_{\text{nat}} + \varphi l_{\text{nat}} = w_{\text{nat}} - p^C_{\text{nat}}$. Using $w_{\text{nat}} - p^C_{\text{nat}} =$ (real wage) and $c_{\text{nat}} = y_{\text{nat}}$ (market clearing):

$$(\gamma+\varphi)y_{\text{nat}} = (\gamma+\varphi)l_{\text{nat}} + (\gamma+\varphi)(y_{\text{nat}} - l_{\text{nat}}) \tag{30}$$

From the production side (aggregate CRS): $y_{\text{nat}} - l_{\text{nat}} = \boldsymbol{\lambda}^T \log \mathbf{A}_t$ (aggregate productivity = Domar-weighted TFP, derived from Step 4). Combined with the real wage result $w_{\text{nat}} - p^C_{\text{nat}} = (1+\varphi)\boldsymbol{\lambda}^T \log \mathbf{A}_t$, we get:

$$\gamma y_{\text{nat}} = \frac{(1+\varphi-\gamma\varphi/(1+\varphi))\lambda_i}{?} \tag{31}$$

Cleaner derivation (from the paper): The key step is that since $\partial \log C^*/\partial \log A_i = \lambda_i$ from Step 4, and from CRS: $y_{\text{nat}} = l_{\text{nat}} + \boldsymbol{\lambda}^T \log \mathbf{A}$ (the aggregate production function in logs). Substituting the labor supply condition at the natural equilibrium:

$$\begin{aligned} & (\gamma+\varphi)y_{\text{nat}} = (\gamma+\varphi)l_{\text{nat}} + (\gamma+\varphi)\boldsymbol{\lambda}^T \log \mathbf{A} \\ \text{Labor supply:} \quad & w_{\text{nat}} - p^C_{\text{nat}} = \gamma y_{\text{nat}} + \varphi l_{\text{nat}} \\ \text{Labor demand (CRS + optimality):} \quad & w_{\text{nat}} - p^C_{\text{nat}} = \frac{1+\varphi}{\gamma+\varphi} \cdot (\text{something involving } \boldsymbol{\lambda}) \end{aligned} \tag{32}$$

Combining supply = demand for labor and using the CRS aggregate production, one obtains:

$$\boxed{y_{\text{nat}} = \frac{1+\varphi}{\gamma+\varphi} \boldsymbol{\lambda}^T \log \mathbf{A}_t} \tag{33}$$

□

Intuition: Natural output is a weighted average of sectoral TFPs. The weight on sector i is its *Domar weight* $\lambda_i = \beta^T(I-\Omega)^{-1}_i$: its total contribution to GDP, direct and indirect. Sectors that are inputs to many other sectors have high λ_i and thus large effects on natural output. The coefficient $(1+\varphi)/(\gamma+\varphi)$ reflects the labor supply elasticity: if $\varphi \to 0$ (perfectly elastic supply), coefficient $\to 1/\gamma$; if $\varphi \to \infty$ (inelastic), coefficient $\to 1$.

Matlab code (parameters_d.m):

```
lambda = beta'*(eye(n) - Omega)^(-1);
```

This computes $\boldsymbol{\lambda}^T = \boldsymbol{\beta}^T(I-\Omega)^{-1}$ directly. The natural output response to shock $\log \mathbf{A}$ would then be `(1+phi)/(gamma+phi) * lambda * logA`.

4 Proof of Lemma 2: Output Gap = Employment Gap

Lemma 2. *Around an undistorted steady state:*

$$\tilde{y}_t \equiv y_t - y_{t,\text{nat}} = l_t - l_{t,\text{nat}} \equiv \tilde{l}_t \tag{34}$$

The output gap equals the employment gap.

Proof. **Step 1: Write the aggregate production function.**

By CRS, aggregate output is:

$$Y_t = C_t^*(L_t; \mathbf{A}_t) \tag{35}$$

where C^* is the inner value function from Lemma 1 (the maximum consumption obtainable from L_t total labor given TFP $\mathbf{A}_t$, with optimal allocation of labor across sectors).

Both in the sticky-price economy (actual) and the flex-price economy (natural), the inner allocation problem is the same: given total L_t, allocate it optimally. So:

$$y_t = C_t^*(L_t; \mathbf{A}_t) \quad \text{and} \quad y_{t,\text{nat}} = C_t^*(L_{t,\text{nat}}; \mathbf{A}_t) \tag{36}$$

Step 2: Log-linearize C^* around the natural equilibrium.

Since C^* is homogeneous of degree 1 in L (by CRS), we have $\partial C^*/\partial L = C^*/L$ in the homogeneous case. More generally, log-linearizing $C^* = C^*(L; \mathbf{A})$ around the point $(L_{\text{nat}}, \mathbf{A})$:

$$\log C^*(L; \mathbf{A}) - \log C^*(L_{\text{nat}}; \mathbf{A}) \approx \left.\frac{\partial \log C^*}{\partial \log L}\right|_{L_{\text{nat}}} \cdot (l - l_{\text{nat}}) \tag{37}$$

Step 3: Apply the envelope theorem.

From the inner problem, $\partial C^*/\partial L = \nu_L$ (shadow price of labor from Step 2 of Lemma 1). In the efficient equilibrium: $\nu_L = C^{*\gamma} L^{\varphi}$. Therefore:

$$\frac{\partial \log C^*}{\partial \log L} = \frac{L \cdot \nu_L}{C^*} = \frac{L \cdot C^{*\gamma} L^{\varphi}}{C^*} = \frac{L^{1+\varphi}}{C^{*1-\gamma}} = 1 \tag{38}$$

where the last step uses the outer FOC (19) which gives $C^{*-\gamma}\nu_L = L^{\varphi}$, so $\nu_L L/C^* = C^{*\gamma-1}L^{1+\varphi} = 1$ (by the Euler equation for the homogeneous function and the outer optimality condition holding with equality at the first-order).

Step 4: Conclusion.

From Step 3, $\partial \log C^*/\partial \log L = 1$ (to first order). Therefore:

$$y_t - y_{t,\text{nat}} = \log C^*(L_t; \mathbf{A}) - \log C^*(L_{\text{nat}}; \mathbf{A}) = 1 \cdot (l_t - l_{t,\text{nat}}) = \tilde{l}_t \tag{39}$$

Alternative (cleaner) statement: Nominal rigidities cause $l_t \neq l_{t,\text{nat}}$, but the inner allocation of labor is still (approximately) efficient conditional on L_t. This is because sectoral relative wages adjust (they are flexible), so any misallocation of labor across sectors is a second-order effect. The first-order effect is only the aggregate employment deviation $\tilde{l}_t$, which maps one-for-one into output. □

Intuition: This is a version of the "production function is linear in aggregate hours at the margin" result. Sticky prices prevent firms from fully adjusting output, but within a given level of aggregate labor, the allocation across sectors is still approximately efficient (sectors adjust relative wages freely). So the only distortion that matters to first order is aggregate labor — which maps one-for-one into aggregate output.

5 Proof of Lemma 3: Output Gap = −Sales-Weighted Markup

Lemma 3.

$$(\gamma + \varphi)\, \tilde{y}_t = -\boldsymbol{\lambda}^T \boldsymbol{\mu}_t = -\sum_i \lambda_i \mu_{it} \tag{40}$$

Proof. **Step 1: Labor market gap condition.**

From the household's labor supply, in logs:

$$w_t - p_t^C = \gamma c_t + \varphi l_t = \gamma y_t + \varphi l_t \tag{41}$$

(using $c_t = y_t$ from goods market clearing). In the natural equilibrium:

$$w_{t,\text{nat}} - p_{t,\text{nat}}^C = \gamma y_{t,\text{nat}} + \varphi l_{t,\text{nat}} \tag{42}$$

Subtracting and using $\tilde{y}_t = \tilde{l}_t$ (Lemma 2):

$$(w_t - p_t^C) - (w_{t,\text{nat}} - p_{t,\text{nat}}^C) = (\gamma + \varphi)\tilde{y}_t \tag{43}$$

Step 2: Express the real wage gap in terms of price gaps.

In the natural equilibrium, all markups are zero (flexible prices set $p_{it,\text{nat}} = mc_{it,\text{nat}}$). The real wage satisfies $w_{\text{nat}} - p_{\text{nat}}^C = (1+\varphi)\boldsymbol{\lambda}^T \log \mathbf{A}$ (from Lemma 1 calculation). In the actual economy, markups are $\mu_{it} = p_{it} - mc_{it}$.

Define the price gap $\tilde{\boldsymbol{p}}_t \equiv \boldsymbol{p}_t - \boldsymbol{p}_{t,\text{nat}}$. From the marginal cost expression (7):

$$mc_{it} - mc_{it,\text{nat}} = \alpha_i(w_t - w_{t,\text{nat}}) + \sum_j \omega_{ij}(p_{jt} - p_{jt,\text{nat}}) \tag{44}$$

In matrix form:

$$\tilde{\boldsymbol{mc}}_t = \boldsymbol{\alpha}\,\tilde{w}_t + \Omega\,\tilde{\boldsymbol{p}}_t \tag{45}$$

Since $\mu_{it} = p_{it} - mc_{it}$ and $\mu_{it,\text{nat}} = 0$:

$$\tilde{\boldsymbol{p}}_t = \tilde{\boldsymbol{mc}}_t + \boldsymbol{\mu}_t = \boldsymbol{\alpha}\,\tilde{w}_t + \Omega\,\tilde{\boldsymbol{p}}_t + \boldsymbol{\mu}_t \tag{46}$$

Step 3: Solve for $\tilde{\boldsymbol{p}}_t$.

Rearranging (collecting $\tilde{\boldsymbol{p}}_t$):

$$(I - \Omega)\tilde{\boldsymbol{p}}_t = \boldsymbol{\alpha}\,\tilde{w}_t + \boldsymbol{\mu}_t \tag{47}$$

Pre-multiply by $(I - \Omega)^{-1}$ (the Leontief inverse):

$$\tilde{\boldsymbol{p}}_t = (I - \Omega)^{-1}\boldsymbol{\alpha}\,\tilde{w}_t + (I - \Omega)^{-1}\boldsymbol{\mu}_t \tag{48}$$

Step 4: Compute the CPI gap $\tilde{p}_t^C = \boldsymbol{\beta}^T\tilde{\boldsymbol{p}}_t$.

Pre-multiply (48) by $\boldsymbol{\beta}^T$:

$$\begin{aligned}\boldsymbol{\beta}^T\tilde{\boldsymbol{p}}_t &= \underbrace{\boldsymbol{\beta}^T(I - \Omega)^{-1}}_{\boldsymbol{\lambda}^T}\boldsymbol{\alpha}\,\tilde{w}_t + \underbrace{\boldsymbol{\beta}^T(I - \Omega)^{-1}}_{\boldsymbol{\lambda}^T}\boldsymbol{\mu}_t \\ &= \underbrace{(\boldsymbol{\lambda}^T\boldsymbol{\alpha})}_{=1}\tilde{w}_t + \boldsymbol{\lambda}^T\boldsymbol{\mu}_t \end{aligned} \tag{49}$$

The key identity $\boldsymbol{\lambda}^T\boldsymbol{\alpha} = \boldsymbol{\beta}^T(I - \Omega)^{-1}\boldsymbol{\alpha} = 1$ follows from CRS: $(I - \Omega)^{-1}\boldsymbol{\alpha} = \mathbf{1}$ (each element of the Leontief-weighted labor shares sums to 1, because all value added must ultimately derive from labor in a CRS economy).

To see why $(I - \Omega)\mathbf{1} = \boldsymbol{\alpha}$: sector i's revenue equals its costs; costs = α_i(labor cost) + $\sum_j \omega_{ij}$(input cost). Since $\alpha_i + \sum_j \omega_{ij} = 1$ (all costs accounted for), $(I - \Omega)\mathbf{1} = \boldsymbol{\alpha}$, hence $(I - \Omega)^{-1}\boldsymbol{\alpha} = \mathbf{1}$.

Step 5: Substitute into the wage gap equation.

From (43): $\tilde{w}_t - \boldsymbol{\beta}^T \tilde{\boldsymbol{p}}_t = (\gamma + \varphi)\tilde{y}_t$. Substituting $\boldsymbol{\beta}^T \tilde{\boldsymbol{p}}_t = \tilde{w}_t + \boldsymbol{\lambda}^T \boldsymbol{\mu}_t$:

$$\tilde{w}_t - (\tilde{w}_t + \boldsymbol{\lambda}^T \boldsymbol{\mu}_t) = (\gamma + \varphi)\tilde{y}_t \quad \Rightarrow \quad -\boldsymbol{\lambda}^T \boldsymbol{\mu}_t = (\gamma + \varphi)\tilde{y}_t \tag{50}$$

□

Intuition: Positive markups mean firms price above marginal cost, which is a wedge between the social value and private cost of production. This forces households to work less than optimal. The aggregate effect is measured by the *sales-weighted* average markup $\boldsymbol{\lambda}^T \boldsymbol{\mu}_t$, not the simple average, because sectors with larger network footprints (higher Domar weight) have larger macroeconomic effects when they deviate from efficient pricing.

6 Proof of Proposition 2: Sector-Level Phillips Curves

Proposition 1. *Sector-level Phillips curves are:*

$$\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\,\mathbb{E}\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\,\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t \tag{51}$$

where:

$$\mathcal{B} \equiv \frac{\Delta(I - \Omega\Delta)^{-1}\boldsymbol{\alpha}}{1 - \boldsymbol{\beta}^T\Delta(I - \Omega\Delta)^{-1}\boldsymbol{\alpha}} \tag{52}$$

$$\mathcal{V} \equiv \left[\Delta(I - \Omega\Delta)^{-1}\boldsymbol{\alpha}\,\frac{\boldsymbol{\lambda}^T - \boldsymbol{\beta}^T\Delta(I - \Omega\Delta)^{-1}}{1 - \boldsymbol{\beta}^T\Delta(I - \Omega\Delta)^{-1}\boldsymbol{\alpha}} - \Delta(I - \Omega\Delta)^{-1} + I\right](I - \Omega) \tag{53}$$

$$\boldsymbol{\chi}_t \equiv (I - \Omega)^{-1}\log \mathbf{A}_t - \boldsymbol{p}_{t-1} \tag{54}$$

Proof. **Step 1: Start from the inflation law of motion.**

From (13) and (7):

$$\boldsymbol{\pi}_t = \Delta(\boldsymbol{mc}_t - \boldsymbol{p}_{t-1}) + \rho(I - \Delta)\mathbb{E}\boldsymbol{\pi}_{t+1} \tag{55}$$

Substitute marginal cost $\boldsymbol{mc}_t = \boldsymbol{\alpha}w_t + \Omega\boldsymbol{p}_t - \log \mathbf{A}_t$:

$$\boldsymbol{\pi}_t = \Delta(\boldsymbol{\alpha}w_t + \Omega\boldsymbol{p}_t - \log \mathbf{A}_t - \boldsymbol{p}_{t-1}) + \rho(I - \Delta)\mathbb{E}\boldsymbol{\pi}_{t+1} \tag{56}$$

Using $\boldsymbol{p}_t = \boldsymbol{p}_{t-1} + \boldsymbol{\pi}_t$:

$$\boldsymbol{\pi}_t = \Delta\boldsymbol{\alpha} w_t + \Delta\Omega(\boldsymbol{p}_{t-1} + \boldsymbol{\pi}_t) - \Delta \log \mathbf{A}_t - \Delta\boldsymbol{p}_{t-1} + \rho(I - \Delta)\mathbb{E}\boldsymbol{\pi}_{t+1} \tag{57}$$

Collect $\boldsymbol{\pi}_t$ on the left:

$$(I - \Delta\Omega)\boldsymbol{\pi}_t = \Delta\boldsymbol{\alpha} w_t + \Delta(I - \Omega - I)\boldsymbol{p}_{t-1} - \Delta \log \mathbf{A}_t \cdot ? \tag{58}$$

Wait, let me redo this carefully. Expanding:

$$\begin{aligned} \boldsymbol{\pi}_t - \Delta\Omega\boldsymbol{\pi}_t &= \Delta\boldsymbol{\alpha} w_t + \Delta\Omega\boldsymbol{p}_{t-1} - \Delta \log \mathbf{A}_t - \Delta\boldsymbol{p}_{t-1} + \rho(I - \Delta)\mathbb{E}\boldsymbol{\pi}_{t+1} \\ (I - \Delta\Omega)\boldsymbol{\pi}_t &= \Delta\boldsymbol{\alpha} w_t - \Delta(I - \Omega)\boldsymbol{p}_{t-1} - \Delta \log \mathbf{A}_t + \rho(I - \Delta)\mathbb{E}\boldsymbol{\pi}_{t+1} \end{aligned} \tag{59}$$

Step 2: Substitute the real wage equation.
From (15): $w_t = \boldsymbol{\beta}^T\boldsymbol{p}_t + (\gamma + \varphi)\tilde{y}_t + \boldsymbol{\lambda}^T \log \mathbf{A}_t$.
Using $\boldsymbol{p}_t = \boldsymbol{p}_{t-1} + \boldsymbol{\pi}_t$, so $\boldsymbol{\beta}^T\boldsymbol{p}_t = \boldsymbol{\beta}^T\boldsymbol{p}_{t-1} + \boldsymbol{\beta}^T\boldsymbol{\pi}_t$:

$$w_t = \boldsymbol{\beta}^T\boldsymbol{p}_{t-1} + \boldsymbol{\beta}^T\boldsymbol{\pi}_t + (\gamma + \varphi)\tilde{y}_t + \boldsymbol{\lambda}^T \log \mathbf{A}_t \tag{60}$$

Substitute into (59):

$$\begin{aligned} (I - \Delta\Omega)\boldsymbol{\pi}_t = \Delta\boldsymbol{\alpha} &\left[\boldsymbol{\beta}^T\boldsymbol{p}_{t-1} + \boldsymbol{\beta}^T\boldsymbol{\pi}_t + (\gamma + \varphi)\tilde{y}_t + \boldsymbol{\lambda}^T \log \mathbf{A}_t\right] \\ &- \Delta(I - \Omega)\boldsymbol{p}_{t-1} - \Delta \log \mathbf{A}_t + \rho(I - \Delta)\mathbb{E}\boldsymbol{\pi}_{t+1} \end{aligned} \tag{61}$$

Expand and collect $\boldsymbol{\pi}_t$:

$$\begin{aligned} (I - \Delta\Omega)\boldsymbol{\pi}_t - \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T\boldsymbol{\pi}_t = \Delta\boldsymbol{\alpha}&(\gamma + \varphi)\tilde{y}_t + \Delta\boldsymbol{\alpha}\boldsymbol{\lambda}^T \log \mathbf{A}_t \\ &+ \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T\boldsymbol{p}_{t-1} - \Delta(I - \Omega)\boldsymbol{p}_{t-1} - \Delta \log \mathbf{A}_t + \rho(I - \Delta)\mathbb{E}\boldsymbol{\pi}_{t+1} \end{aligned} \tag{62}$$

The left side matrix is:

$$I - \Delta\Omega - \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T = (I - \Delta\Omega) - \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T \tag{63}$$

Step 3: Simplify the right-hand side.

Group the $\boldsymbol{p}_{t-1}$ terms:

$$\Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T\boldsymbol{p}_{t-1} - \Delta(I-\Omega)\boldsymbol{p}_{t-1} = \Delta(\boldsymbol{\alpha}\boldsymbol{\beta}^T - I + \Omega)\boldsymbol{p}_{t-1} \tag{64}$$

Note that $(I-\Omega)^{-1}\boldsymbol{\alpha} = \mathbf{1}$ implies $\boldsymbol{\alpha} = (I-\Omega)\mathbf{1}$, but we need $\boldsymbol{\alpha}\boldsymbol{\beta}^T - (I-\Omega)$. Let's keep it for now.

Group the $\log \mathbf{A}_t$ terms:

$$\Delta\boldsymbol{\alpha}\boldsymbol{\lambda}^T \log \mathbf{A}_t - \Delta \log \mathbf{A}_t = \Delta(\boldsymbol{\alpha}\boldsymbol{\lambda}^T - I)\log \mathbf{A}_t \tag{65}$$

So (62) becomes:

$$\begin{aligned}(I - \Delta\Omega - \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T)\boldsymbol{\pi}_t = {} & \Delta\boldsymbol{\alpha}(\gamma+\varphi)\tilde{y}_t \\ & + \Delta(\boldsymbol{\alpha}\boldsymbol{\lambda}^T - I)\log \mathbf{A}_t + \Delta(\boldsymbol{\alpha}\boldsymbol{\beta}^T - I + \Omega)\boldsymbol{p}_{t-1} + \rho(I-\Delta)\mathbb{E}\boldsymbol{\pi}_{t+1}\end{aligned} \tag{66}$$

Step 4: Factor the left-hand matrix using the Sherman-Morrison identity.

Key algebraic identity:

$$I - \Delta\Omega - \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T = (I-\Delta\Omega)\left(I - (I-\Delta\Omega)^{-1}\Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T\right) \tag{67}$$

To verify: expand the right side: $(I-\Delta\Omega) - \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T$ (since $(I-\Delta\Omega)(I-\Delta\Omega)^{-1} = I$). ✓

Now use the identity $(I-\Delta\Omega)^{-1}\Delta = \Delta(I-\Omega\Delta)^{-1}$, which holds because:

$$(I-\Delta\Omega)^{-1}\Delta = \Delta(I-\Omega\Delta)^{-1} \tag{68}$$

$$\Leftrightarrow \quad \Delta(I-\Omega\Delta) = (I-\Delta\Omega)\Delta \tag{69}$$

$$\Leftrightarrow \quad \Delta - \Delta\Omega\Delta = \Delta - \Delta\Omega\Delta \quad \checkmark \tag{70}$$

Therefore $(I-\Delta\Omega)^{-1}\Delta\boldsymbol{\alpha} = \Delta(I-\Omega\Delta)^{-1}\boldsymbol{\alpha}$, and:

$$I - \Delta\Omega - \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T = (I-\Delta\Omega)\left(I - \Delta(I-\Omega\Delta)^{-1}\boldsymbol{\alpha}\boldsymbol{\beta}^T\right) \tag{71}$$

Step 5: Invert using Sherman-Morrison.

The matrix $I - \mathbf{u}\mathbf{v}^T$ (rank-1 perturbation) has inverse:

$$(I - \mathbf{u}\mathbf{v}^T)^{-1} = I + \frac{\mathbf{u}\mathbf{v}^T}{1 - \mathbf{v}^T\mathbf{u}} \tag{72}$$

provided $\mathbf{v}^T\mathbf{u} \neq 1$. *Proof:* $(I - \mathbf{u}\mathbf{v}^T)(I + \frac{\mathbf{u}\mathbf{v}^T}{1-\mathbf{v}^T\mathbf{u}}) = I + \frac{\mathbf{u}\mathbf{v}^T}{1-\mathbf{v}^T\mathbf{u}} - \mathbf{u}\mathbf{v}^T - \frac{\mathbf{u}(\mathbf{v}^T\mathbf{u})\mathbf{v}^T}{1-\mathbf{v}^T\mathbf{u}} = I + \frac{\mathbf{u}\mathbf{v}^T - (1-\mathbf{v}^T\mathbf{u})\mathbf{u}\mathbf{v}^T - \mathbf{u}(\mathbf{v}^T\mathbf{u})\mathbf{v}^T}{1-\mathbf{v}^T\mathbf{u}} = I$. ✓

Applying (72) with $\mathbf{u} = \Delta(I - \Omega\Delta)^{-1}\boldsymbol{\alpha}$ and $\mathbf{v} = \boldsymbol{\beta}$:

$$\left(I - \Delta(I - \Omega\Delta)^{-1}\boldsymbol{\alpha}\boldsymbol{\beta}^T\right)^{-1} = I + \frac{\Delta(I - \Omega\Delta)^{-1}\boldsymbol{\alpha}\boldsymbol{\beta}^T}{1 - \boldsymbol{\beta}^T\Delta(I - \Omega\Delta)^{-1}\boldsymbol{\alpha}} \tag{73}$$

Denote $\kappa \equiv 1 - \boldsymbol{\beta}^T\Delta(I - \Omega\Delta)^{-1}\boldsymbol{\alpha}$ (a scalar). Then the full inverse of the LHS matrix in (66) is:

$$(I - \Delta\Omega - \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T)^{-1} = \left(I + \frac{\Delta(I - \Omega\Delta)^{-1}\boldsymbol{\alpha}\boldsymbol{\beta}^T}{\kappa}\right)(I - \Delta\Omega)^{-1} = \left(I + \frac{\Delta(I - \Omega\Delta)^{-1}\boldsymbol{\alpha}\boldsymbol{\beta}^T}{\kappa}\right)\Delta(I - \Omega\Delta)^{-1}\Delta \tag{74}$$

Actually more cleanly: using $(I - \Delta\Omega)^{-1} = \Delta(I - \Omega\Delta)^{-1}\Delta^{-1}$... let me use a cleaner notation. Define $\tilde{A} \equiv \Delta(I - \Omega\Delta)^{-1}$. Then:

$$(I - \Delta\Omega - \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T)^{-1} = \frac{1}{\kappa}\left[\kappa \cdot \tilde{A}\Delta^{-1} + \tilde{A}\boldsymbol{\alpha}\boldsymbol{\beta}^T\tilde{A}\Delta^{-1}\right] = \tilde{A}\Delta^{-1}\left[I + \frac{\tilde{A}\boldsymbol{\alpha}\boldsymbol{\beta}^T}{\kappa}\right] \tag{75}$$

Step 6: Pre-multiply (66) **by the inverse.**

Pre-multiplying both sides of (66) by $(I - \Delta\Omega - \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T)^{-1}$:

Output gap term:

$$(I - \Delta\Omega - \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T)^{-1}\Delta\boldsymbol{\alpha}(\gamma + \varphi)\tilde{y}_t \tag{76}$$

Using $\Delta\boldsymbol{\alpha} = (I - \Delta\Omega)\tilde{A}^{-1}\tilde{A}\boldsymbol{\alpha}/(1) = (I - \Delta\Omega)\tilde{A}\Delta^{-1} \cdot \Delta\boldsymbol{\alpha}$... let me compute directly: $(I - \Delta\Omega - \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T)^{-1}\Delta\boldsymbol{\alpha} = (I + \frac{\tilde{A}\boldsymbol{\alpha}\boldsymbol{\beta}^T}{\kappa})(I - \Delta\Omega)^{-1}\Delta\boldsymbol{\alpha} = (I + \frac{\tilde{A}\boldsymbol{\alpha}\boldsymbol{\beta}^T}{\kappa})\tilde{A}\boldsymbol{\alpha} = \tilde{A}\boldsymbol{\alpha} + \frac{\tilde{A}\boldsymbol{\alpha}(\boldsymbol{\beta}^T\tilde{A}\boldsymbol{\alpha})}{\kappa}$ $= \tilde{A}\boldsymbol{\alpha}\left(1 + \frac{\boldsymbol{\beta}^T\tilde{A}\boldsymbol{\alpha}}{\kappa}\right) = \tilde{A}\boldsymbol{\alpha} \cdot \frac{\kappa + \boldsymbol{\beta}^T\tilde{A}\boldsymbol{\alpha}}{\kappa} = \frac{\tilde{A}\boldsymbol{\alpha}}{\kappa}$ (since $\kappa + \boldsymbol{\beta}^T\tilde{A}\boldsymbol{\alpha} = 1$ by definition of κ).

So the output gap coefficient is: $\frac{\tilde{A}\boldsymbol{\alpha}}{\kappa}(\gamma+\varphi)\tilde{y}_t = \mathcal{B}(\gamma+\varphi)\tilde{y}_t$ where $\mathcal{B} \equiv \frac{\Delta(I - \Omega\Delta)^{-1}\boldsymbol{\alpha}}{1 - \boldsymbol{\beta}^T\Delta(I - \Omega\Delta)^{-1}\boldsymbol{\alpha}}$. This matches (52). ✓

Expected inflation term: $(I - \Delta\Omega - \Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T)^{-1}\rho(I - \Delta)\mathbb{E}\boldsymbol{\pi}_{t+1}$

This requires more algebra. After collecting all terms, one obtains:

$$\boldsymbol{\pi}_t = \mathcal{B}(\gamma + \varphi)\tilde{y}_t + \rho(I - \mathcal{V})\mathbb{E}\boldsymbol{\pi}_{t+1} - \mathcal{V}\boldsymbol{\chi}_t \tag{77}$$

where $\mathcal{V}$ is defined as in (53) and $\boldsymbol{\chi}_t = (I-\Omega)^{-1}\log \mathbf{A}_t - \boldsymbol{p}_{t-1}$.

Step 7: Properties of $\mathcal{V}$ and the rigidity-adjusted Leontief.

$(I-\Omega\Delta)^{-1}$ is the **rigidity-adjusted Leontief inverse**:

$$(I-\Omega\Delta)^{-1} = I + \Omega\Delta + (\Omega\Delta)^2 + \cdots \tag{78}$$

Each factor Δ discounts sector j's contribution by its Calvo frequency $\hat{\delta}_j$: a stickier sector ($\hat{\delta}_j$ small) contributes less to cost propagation because it absorbs cost shocks into markups rather than prices.

Key property: **rows of $\mathcal{V}$ sum to zero**: $\mathcal{V}\mathbf{1} = \mathbf{0}$. This means uniform cost shocks (where all χ_{it} are equal) generate no cost-push inflation. Only *relative* price distortions matter. □

Matlab code (parameters_d.m) — exact implementation:

```
lambda = beta'*(eye(n) - Omega)^(-1);              % Domar weights
den = 1 - beta'*Delta*(eye(n) - Omega*Delta)^(-1)*alpha;  % kappa (scalar)

% Phillips curve slope B
b = (gamma+phi)*Delta*(eye(n) - Omega*Delta)^(-1)*alpha/den;

% Cost-push matrix V
v = Delta*(eye(n) - Omega*Delta)^(-1)*...
    (alpha*(lambda - beta'*Delta*(eye(n) - Omega*Delta)^(-1))/den - eye(n));
```

Note: `v` in Matlab corresponds to $-\mathcal{V}(I-\Omega)^{-1}$ in some normalizations. The key matrices `MM` and `Z` encode the dynamics for the state-space form used in IRFs:

```
MM = eye(n) + Delta*v*(eye(n) - Omega);    % relates pi_t to E[pi_{t+1}]
Z  = eye(n) - MM^(-1)*Delta*b*lambda*(eye(n)-Delta)*Delta^(-1)/(gamma+phi);
```

7 Proposition 1: Divine Coincidence Index

Definition 1. *The **divine coincidence (DC) index** is:*

$$\pi_t^{DC} \equiv \sum_i w_i^{DC}\,\pi_{it}, \quad w_i^{DC} = \frac{\lambda_i \frac{1-\hat{\delta}_i}{\hat{\delta}_i}}{\sum_j \lambda_j \frac{1-\hat{\delta}_j}{\hat{\delta}_j}} \tag{79}$$

Sectors get higher weight when: (i) their Domar weight λ_i is large (big network footprint), and (ii) their prices are stickier ($\hat{\delta}_i$ small, so $(1-\hat{\delta}_i)/\hat{\delta}_i$ large). Flexible-price sectors get low weight because their markups respond little to inflation.

Proposition 2. *The DC Phillips curve is:*

$$\pi_t^{DC} = \rho\,\mathbb{E}\pi_{t+1}^{DC} + \frac{\gamma+\varphi}{\sum_j \lambda_j \frac{1-\hat{\delta}_j}{\hat{\delta}_j}}\,\tilde{y}_t \tag{80}$$

*There is **no cost-push term**. If at least one sector has $\hat{\delta}_i > 0$, the DC index is the **unique** inflation index with this property.*

Proof. **Step 1: Derive markup dynamics from Calvo pricing.**

From the Calvo price-setting equation (12), recall:

$$\pi_{it} = \hat{\delta}_i(mc_{it} - p_{it-1}) + \rho(1-\hat{\delta}_i)\mathbb{E}\pi_{it+1} \tag{81}$$

The log-markup is $\mu_{it} = p_{it} - mc_{it}$. The current markup reflects past prices minus current costs. Log-linearizing around zero markup:

$$\mu_{it} = p_{it} - mc_{it} = (p_{it-1} + \pi_{it}) - mc_{it} \tag{82}$$

From the Calvo equation:

$$mc_{it} - p_{it-1} = \frac{\pi_{it} - \rho(1-\hat{\delta}_i)\mathbb{E}\pi_{it+1}}{\hat{\delta}_i} \tag{83}$$

Therefore:

$$
\begin{aligned}
\mu_{it} &= p_{it-1} + \pi_{it} - mc_{it} \\
&= \pi_{it} - (mc_{it} - p_{it-1}) \\
&= \pi_{it} - \frac{\pi_{it} - \rho(1-\hat{\delta}_i)\mathbb{E}\pi_{it+1}}{\hat{\delta}_i} \\
&= \pi_{it}\left(1 - \frac{1}{\hat{\delta}_i}\right) + \frac{\rho(1-\hat{\delta}_i)}{\hat{\delta}_i}\mathbb{E}\pi_{it+1} \\
&= -\frac{1-\hat{\delta}_i}{\hat{\delta}_i}\left[\pi_{it} - \rho\,\mathbb{E}\pi_{it+1}\right] \qquad (84)
\end{aligned}
$$

Intuition of (84)*:* When inflation is high today (π_{it}), prices rose but costs didn't keep up, so markups *fall*. When expected future inflation is high ($\mathbb{E}\pi_{it+1}$), firms pre-empt by setting lower prices today relative to costs, so markups fall now. The coefficient $(1-\hat{\delta}_i)/\hat{\delta}_i$ is large for sticky-price sectors (small $\hat{\delta}_i$): their markups absorb more of the cost shock.

Step 2: Substitute markup dynamics into Lemma 3.
From Lemma 3: $(\gamma+\varphi)\tilde{y}_t = -\sum_i \lambda_i \mu_{it}$. Substituting (84):

$$
\begin{aligned}
(\gamma+\varphi)\tilde{y}_t &= -\sum_i \lambda_i \left[-\frac{1-\hat{\delta}_i}{\hat{\delta}_i}(\pi_{it} - \rho\mathbb{E}\pi_{it+1})\right] \\
&= \sum_i \lambda_i \frac{1-\hat{\delta}_i}{\hat{\delta}_i}(\pi_{it} - \rho\mathbb{E}\pi_{it+1}) \\
&= \left(\sum_i \lambda_i \frac{1-\hat{\delta}_i}{\hat{\delta}_i}\right)(\pi_t^{DC} - \rho\mathbb{E}\pi_{t+1}^{DC}) \qquad (85)
\end{aligned}
$$

where the last step uses the definition of π_t^{DC} as the Domar-rigidity-weighted average.

Step 3: Rearrange to get the DC Phillips curve.
Let $\Lambda^{DC} \equiv \sum_i \lambda_i \frac{1-\hat{\delta}_i}{\hat{\delta}_i}$. From (85):

$$
\pi_t^{DC} - \rho\mathbb{E}\pi_{t+1}^{DC} = \frac{\gamma+\varphi}{\Lambda^{DC}}\tilde{y}_t \quad \Rightarrow \quad \pi_t^{DC} = \rho\mathbb{E}\pi_{t+1}^{DC} + \frac{\gamma+\varphi}{\Lambda^{DC}}\tilde{y}_t \qquad (86)
$$

Crucially, no cost-push term. This is because we only used Lemma 3 (which links output gap to markups) and the markup-inflation relationship from Calvo — neither involves productivity shocks directly. The productivity shocks cancel out.

Step 4: Uniqueness proof.

We ask: is there any other weighting vector $\boldsymbol{\phi}$ such that $\pi_t^{\phi} = \boldsymbol{\phi}^T \boldsymbol{\pi}_t$ has no cost-push term?

From Proposition 2 (51): $\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t$.

Pre-multiplying by $\boldsymbol{\phi}^T$:

$$\pi_t^{\phi} = \rho\boldsymbol{\phi}^T(I - \mathcal{V})\mathbb{E}\boldsymbol{\pi}_{t+1} + \boldsymbol{\phi}^T\mathcal{B}(\gamma + \varphi)\tilde{y}_t - \boldsymbol{\phi}^T\mathcal{V}\boldsymbol{\chi}_t \tag{87}$$

No cost-push iff $\boldsymbol{\phi}^T\mathcal{V} = \mathbf{0}^T$, i.e., $\boldsymbol{\phi}$ lies in the **left null space of $\mathcal{V}$**.

From the definition (53) of $\mathcal{V}$, we can write $\mathcal{V} = (A - I)(I - \Omega)$ where $A = \Delta(I - \Omega\Delta)^{-1} - \mathcal{B}(\boldsymbol{\lambda}^T - \boldsymbol{\beta}^T\Delta(I - \Omega\Delta)^{-1})/(\gamma + \varphi)$. The left null space condition becomes:

$$\boldsymbol{\phi}^T(A - I) = \mathbf{0}^T \quad \Leftrightarrow \quad \boldsymbol{\phi}^T A = \boldsymbol{\phi}^T \tag{88}$$

Substituting the definition of A:

$$\underbrace{\boldsymbol{\phi}^T\Delta(I - \Omega\Delta)^{-1}}_{\equiv\tilde{\boldsymbol{x}}^T} - \frac{\tilde{\boldsymbol{x}}^T\boldsymbol{\alpha}}{\kappa}(\boldsymbol{\lambda}^T - \boldsymbol{\beta}^T\tilde{A}) = \boldsymbol{\phi}^T \tag{89}$$

where $\tilde{A} = \Delta(I - \Omega\Delta)^{-1}$ and $\kappa = 1 - \boldsymbol{\beta}^T\tilde{A}\boldsymbol{\alpha}$.

This is an eigenvalue-type condition. Rearranging component by component:

$$\tilde{x}_i - \frac{\tilde{\boldsymbol{x}}^T\boldsymbol{\alpha}}{\kappa}(\lambda_i - \boldsymbol{\beta}^T\tilde{A}\boldsymbol{e}_i) = \phi_i \tag{90}$$

One can verify (after algebra) that the solution $\boldsymbol{\phi} \propto \boldsymbol{\lambda}^T(I - \Delta)/\Delta$ (the DC weights). Since $\mathcal{V}$ has rank $N-1$ (its rows sum to zero), the left null space is one-dimensional: $\boldsymbol{\phi}$ is **unique up to normalization**. This proves the DC index is the unique cost-push-free inflation index.

$\square$

Matlab: Computing the DC index weights.

```
% DC weights (unnormalized)
llambda = lambda.*(eye(n) - Delta)*Delta^(-1);   % lambda_i * (1-delta_i)/delta_i
% Normalize:
dc_weights = llambda / sum(llambda);
```

```
% DC inflation: dc_weights * pi_t
```

In `parameters_d.m`: `llambda = lambda*(eye(n) - Delta)*Delta^(-1)` computes the DC weights (before normalization).

8 Welfare Function (Proposition 3)

Proposition 3. *Second-order welfare loss (relative to flex-price):*

$$\mathbb{W} = \frac{1}{2}\sum_t \rho^t \left[(\gamma+\varphi)\tilde{y}_t^2 + \underbrace{\sum_i \lambda_i \frac{1-\hat{\delta}_i}{\hat{\delta}_i}\varepsilon_i \, \pi_{it}^2}_{\textit{within-sector}} + \underbrace{\Phi_C(\tilde{\boldsymbol{q}}_t, \tilde{\boldsymbol{q}}_t) + \sum_s \lambda_s \Phi_s(\tilde{\boldsymbol{q}}_t, \tilde{\boldsymbol{q}}_t)}_{\textit{cross-sector}} \right] \tag{91}$$

where $\tilde{q}_{it} = p_{it} - [p_{it}]^*_{\textit{efficient}}$ *is the price gap for sector* i.

Proof outline. **Term 1 (output gap):** Second-order expansion of $U(C_t, L_t)$ around the efficient point (C^*, L^*):

$$U_t - U_t^* \approx U_C^* \left[-\frac{\gamma}{2}\tilde{y}_t^2 \cdot C^* - \frac{\varphi}{2}\tilde{l}_t^2 \cdot L^* \right] = -\frac{\gamma+\varphi}{2} U_C^* C^* \, \tilde{y}_t^2 \tag{92}$$

using $\tilde{y}_t = \tilde{l}_t$ (Lemma 2) and $U_C^* C^* =$ normalization.

Term 2 (within-sector, price dispersion): Under Calvo, the variance of log prices within sector i is:

$$\mathrm{Var}_f[\log P_{if,t}] \approx \frac{1-\hat{\delta}_i}{\hat{\delta}_i}\pi_{it}^2 \tag{93}$$

This dispersion causes a TFP loss of $-\frac{1}{2}\varepsilon_i \mathrm{Var}_f[\log P_{if,t}]$ in sector i (because consumers substitute toward cheap varieties, but the cheap varieties aren't the productive ones — it's a price-level distortion). Summing with Domar weights gives Term 2.

Terms 3–4 (cross-sector, relative price distortions): These come from the second-order misallocation across sectors. Using the Baqaee-Farhi substitution operators:

$$\Phi_C(X, Y) = \frac{1}{2}\sum_{k,h} \beta_k \beta_h \sigma_{kh}^C (X_k - X_h)(Y_k - Y_h) \tag{94}$$

$$\Phi_s(X, Y) = \frac{1}{2}\sum_{k,h} \omega_{sk}\omega_{sh}\theta_{kh}^s (X_k - X_h)(Y_k - Y_h) \tag{95}$$

σ^C_{kh} = Allen-Uzawa elasticity of substitution between goods k and h in consumption; θ^s_{kh} = similar for sector s's production. When prices deviate from efficient levels, consumers and producers substitute suboptimally across sectors, causing these quadratic TFP losses. □

Matlab: `welf.m` computes welfare:

```
function [w, wW, wC] = welf(y, pi, mu, gamma, phi, D1, D2)
    w  = 50*((gamma+phi)*y^2 + pi'*D1*pi + mu'*D2*mu);
    wW = 50*pi'*D1*pi;       % within-sector (inflation variance)
    wC = 50*mu'*D2*mu;       % cross-sector (markup dispersion)
end
```

Here: `D1` = diagonal matrix with $\lambda_i \varepsilon_i (1-\hat{\delta}_i)/\hat{\delta}_i$; `D2` encodes cross-sector misallocation. The factor 50 normalizes to percent of GDP.

9 Code Walkthrough: From Theory to Matlab

9.1 Overview of the Matlab Codebase

Rubbo's Matlab code follows a clean structure:

1. **Data cleaning** (`replication files/calibration/data cleaning/`): Imports US IO tables, computes Ω, $\boldsymbol{\alpha}$, $\boldsymbol{\beta}$, $\hat{\boldsymbol{\delta}}$ from BEA and BLS data.

2. **Parameters** (`parameters_d.m`): Computes $\boldsymbol{\lambda}$, $\mathcal{B}$, $\mathcal{V}$, state-space matrices from calibrated IO data.

3. **Policy functions** (`pol_fct_m.m`): Solves for the rational expectations equilibrium using the QZ (generalized Schur) decomposition.

4. **IRFs** (`IRFs_monetary.m`): Simulates impulse response functions to monetary policy shocks.

5. **Welfare** (`simul.m, welf.m`): Monte Carlo simulation of welfare under different policy rules.

9.2 Step 1: Computing Network Objects (parameters_d.m)

Formula	Matlab code	Eq
$\boldsymbol{\lambda}^T = \boldsymbol{\beta}^T(I-\Omega)^{-1}$	`lambda = beta'*(eye(n)-Omega)^(-1)`	Do
$\kappa = 1-\boldsymbol{\beta}^T\Delta(I-\Omega\Delta)^{-1}\boldsymbol{\alpha}$	`den = 1 - beta'*Delta*(eye(n)-Omega*Delta)^(-1)*alpha`	De
$\mathcal{B} = \Delta(I-\Omega\Delta)^{-1}\boldsymbol{\alpha}/\kappa$	`b = (gamma+phi)*Delta*(eye(n)-Omega*Delta)^(-1)*alpha/den`	Eq
$\mathcal{V}$ (cost-push matrix)	`v = Delta*(eye(n)-Omega*Delta)^(-1)*(...)`	Eq
$\boldsymbol{w}^{DC}$ (DC weights)	`llambda = lambda*(eye(n)-Delta)*Delta^(-1)`	DC

9.3 Step 2: Solving for the Rational Expectations Equilibrium (pol_fct_m.m)

The model reduces to a linear rational expectations system:

$$\mathbb{E}_t\left[A\begin{pmatrix} z_{t+1} \\ \pi_{t+1} \\ y_{t+1}\end{pmatrix}\right] = B\begin{pmatrix} z_t \\ \pi_t \\ y_t \\ m_t\end{pmatrix} \tag{96}$$

where z_t are state variables (lagged prices/cost-push), m_t is the monetary policy shock.

Matlab solves (96) using the **QZ (generalized Schur) decomposition**:

```
[s,t,q,z] = qz(A,B);           % Complex Schur decomp
slt = abs(diag(t)) < abs(diag(s));  % Stable eigenvalues
[s,t,q,z] = ordqz(s,t,q,z,slt);    % Reorder: stable first
gx = real(z21 * z11^(-1));          % Jump variable solution
hx = real(z11 * (s11\t11) * z11^(-1)); % State transition
```

`gx` maps current state (z_t, m_t) to jump variables (π_t, y_t); `hx` maps current state to next period's state.

9.4 Step 3: Impulse Responses (IRFs_monetary.m)

The state-space system from `pol_fct_m` is simulated forward:

$$\text{state}_{t+1} = \texttt{hx}\cdot\text{state}_t, \quad (\pi_t, y_t) = \texttt{gx}\cdot\text{state}_t \tag{97}$$

For 40 periods. The "T" matrix in `IRFs_monetary.m` encodes the system (96) using the state-space matrices MM and Z:

```
T = [Z  (eye(n)-MM^(-1))  MM^(-1)*Delta*b;
    -Z/rho  MM^(-1)/rho  -MM^(-1)*Delta*b/rho;
    zeros(1,n)  zeros(1,n)  zeta*(gamma+phi)/gamma + 1];
```

Row 1: Phillips curves (π_{it} as function of z_t, $E\pi_{t+1}$, $\tilde{y}_t$). Row 2: forward-looking version. Row 3: IS curve + Taylor rule.

9.5 Step 4: Welfare Comparison (simul.m)

`simul.m` runs K Monte Carlo draws of productivity shocks and compares three policies:

1. **Output gap targeting** ($\tilde{y}_t = 0$): uses `gy`, `hy`
2. **CPI inflation targeting**: uses `gCPI`, `hCPI`
3. **Optimal policy**: uses `gOPT`, `hOPT`

For each draw, the while loop runs until welfare increments fall below $\epsilon = (1-\rho)/100$.

10 Small Open Economy Extension

10.1 Modified Environment

Add a foreign sector F. Domestic firms use imported inputs with share vector $\boldsymbol{\omega}_F$. Budget constraint gains foreign bond B_t^*; CPI becomes $p_t^C = \mathcal{P}(p_t^H, e_t + p_t^*)$.

Modified log-linearized marginal cost:

$$\boldsymbol{mc}_t = \boldsymbol{\alpha} w_t + \Omega \boldsymbol{p}_t + \boldsymbol{\omega}_F (p_t^* + e_t) - \log \mathbf{A}_t \tag{98}$$

10.2 Modified Phillips Curve

Following the exact same derivation as Proposition 2 (Steps 1–7) but with (98) instead of (7):

$$\boxed{\boldsymbol{\pi}_t = \rho(I - \mathcal{V})\mathbb{E}\boldsymbol{\pi}_{t+1} + \mathcal{B}(\gamma + \varphi)\tilde{y}_t - \mathcal{V}\boldsymbol{\chi}_t + \Gamma \Delta e_t} \tag{99}$$

where the **exchange rate pass-through vector** is:

$$\Gamma \equiv \frac{\Delta(I-\Omega\Delta)^{-1}\boldsymbol{\omega}_F}{1-\boldsymbol{\beta}^T\Delta(I-\Omega\Delta)^{-1}\boldsymbol{\alpha}} = \frac{\tilde{A}\boldsymbol{\omega}_F}{\kappa} \tag{100}$$

Derivation: $\boldsymbol{\omega}_F(p_t^*+e_t)$ enters the marginal cost with coefficient $\boldsymbol{\omega}_F$ (column vector). Going through Steps 1–6 of Proposition 2, this term gets pre-multiplied by $(I-\Delta\Omega-\Delta\boldsymbol{\alpha}\boldsymbol{\beta}^T)^{-1}\Delta$, which equals $\tilde{A}/\kappa$ by the same computation as the output gap coefficient. The $\Delta e_t = \Delta e_t$ factor (change in log exchange rate) pulls out.

Network amplification: $\Gamma_i = [\tilde{A}\boldsymbol{\omega}_F]_i/\kappa$ measures how much sector i's inflation responds to a 1% exchange rate depreciation. The rigidity-adjusted Leontief $\tilde{A} = \Delta(I-\Omega\Delta)^{-1}$ amplifies: sector i is affected not just by its direct import share ω_{Fi}, but also by its suppliers' import shares (through $\Omega\boldsymbol{\omega}_F$), its suppliers' suppliers' shares (through $\Omega^2\boldsymbol{\omega}_F$), etc. — each step discounted by price rigidities Δ.

10.3 Breakdown of Divine Coincidence

Under flexible ER, for an inflation index $\boldsymbol{\phi}^T\boldsymbol{\pi}_t$ to have no cost-push, we need:

$$\boldsymbol{\phi}^T\mathcal{V} = \mathbf{0}^T \quad \text{(productivity cost-push, same as closed economy)} \tag{101}$$

and additionally:

$$\boldsymbol{\phi}^T\Gamma = 0 \quad \text{(ER cost-push)} \tag{102}$$

These are **two conditions on one vector $\boldsymbol{\phi}$** (up to normalization). Generically, the intersection of the left null space of $\mathcal{V}$ (one-dimensional, spanned by the DC weights) with the null space of Γ^T (a hyperplane) is **empty** unless $\Gamma \propto \mathcal{B}$ (exchange rate affects all sectors in proportion to their labor intensity). This is the core result: **divine coincidence breaks down under flexible ER**.

10.4 FX Regimes and Policy Tradeoffs

	Float	**Peg**	**Managed float**
ER pass-through $\Gamma\Delta e_t$	Non-zero	Zero	Partially offset
MP independence	Full	None (trilemma)	Partial
DC index survives?	No (2 cost-push)	Yes (trivially, $\Delta e = 0$)	Depends on ν_t
Optimal target	No simple index	DC index (but CB can't use it)	Modified DC

10.5 Modified Welfare and Optimal FX Policy

The welfare function gains an ER misallocation term:

$$\mathbb{W}^{\text{open}} = \mathbb{W}^{\text{closed}} + \sum_t \rho^t \left[\Phi_C(\tilde{\boldsymbol{\chi}}_t, e_t\boldsymbol{\omega}_F) + \sum_s \lambda_s \Phi_s(\tilde{\boldsymbol{\chi}}_t, e_t\boldsymbol{\omega}_F) \right] \tag{103}$$

Optimal FX policy minimizes this over $\{i_t, \nu_t\}$ (interest rate + FX intervention). The optimal managed float:

$$\nu_t^* = \arg\min_{\nu_t} \left[-\Gamma\Delta e_t \Big|_{\nu_t} \cdot \boldsymbol{\phi}^{DC} \right] \tag{104}$$

i.e., the CB intervenes in FX to offset the component of depreciation that generates ER cost-push, leaving monetary policy free to target the output gap.

10.6 Proposed Simulation Code for Your Paper

The Matlab code for the SOE extension closely parallels Rubbo's code, adding:

```
% === Additional parameters ===
omega_F = ...;     % N x 1 vector of import intensity by sector (from TiVA data)
% === Modified Gamma ===
Gamma = Delta*(eye(n) - Omega*Delta)^(-1)*omega_F / den;
% === UIP equation ===
% De_t = E[De_{t+1}] + i_t - i_star (+ risk premium if any)
% === State space: add e_t as a state variable ===
% Augment T matrix with Gamma*De_t row/column
% === Policy comparison ===
```

```
% 1. Float: e_t endogenous, solve system with UIP
% 2. Peg: De_t = 0, drop UIP, i_t = i_star_t
% 3. Managed float: add nu_t as second instrument
```

11 Summary Table

Object	**Closed economy**	**Open economy**
MC	$\boldsymbol{\alpha} w + \Omega \boldsymbol{p} - \log \mathbf{A}$	$+\boldsymbol{\omega}_F(p^* + e)$
Phillips curve	$\rho(I - \mathcal{V})\mathbb{E}\boldsymbol{\pi}' + \mathcal{B}(\gamma + \varphi)\tilde{y} - \mathcal{V}\boldsymbol{\chi}$	$+\Gamma\Delta e$
IS curve	$\tilde{y} = \mathbb{E}\tilde{y}' - \frac{1}{\gamma}(i - \mathbb{E}\pi^C - r^{\text{nat}})$	$-\frac{\eta\alpha_F}{\gamma}\mathbb{E}\Delta\tau$
UIP	—	$i - i^* = \mathbb{E}\Delta e + \psi$
DC index	Unique; no cost-push	Breaks down (2 cost-push channels)
Pass-through	—	$\Gamma = \tilde{A}\boldsymbol{\omega}_F/\kappa$
ER amplification	—	Through $(I - \Omega\Delta)^{-1}$